# MIRJAN ALI SHA

*GIS Engineer, Nurture Farm Agtech Pvt. Ltd.*

## PROFILE & OBJECTIVE

GIS Engineer and Data Scientist with 5+ years of experience building satellite data pipelines, ML models, and cloud infrastructure for real-world agricultural applications. An active open-source contributor with 15+ QGIS plugins and 4 PyPI packages totalling 100K+ downloads. Looking to grow at the intersection of Geospatial technology and AI, where I can contribute meaningfully and continue evolving my craft.

**** 2 Months Notice Period (Negotiable).***

## PERSONAL DETAILS

**DATE OF BIRTH:**
12-03-1997

**ADDRESS:**
211, Reliable Pride Apartment, Somasudarapalya, HSR Layout, Bangalore, Karnataka – 560102

**Permanent Address:**
Pratapnagar (Badartala), Arambagh, Hooghly, West Bengal, India - 712415

## CONTACT

PHONE
9091095337 *

* Due to limited cellular reception, please feel free to contact me via WhatsApp or email/G-Chat/MS-Teams anytime; phone calls may not always connect.

**WhatsApp [online almost all day]:**
Chat without save the number:
https://wa.me/9091095337?text=Hi

**Emails and Chats:**
mirjanalisha@gmail.com (G-Chat),
mirjanalisha@outlook.com (MS-Teams)

**ONLINE RESUME (Website):**
https://mirjanalisha.github.io

**LINKEDIN ID:**
https://www.linkedin.com/in/mirjanalisha

## LANGUAGE PROFICIENCY

BENGALI : COLLOQUIAL
HINDI : GOOD
ENGLISH : GOOD

## EDUCATION

**VIDYASAGAR UNIVERSITY,** *M.Sc. in Remote Sensing and GIS*
2018 – 2020 CGPA: 8.44

**IBM Skills Network (Cousera),** *Professional Certificate in Data Science and Machine Learning*
2019 – 2020 9 Courses

**IIIT – Bangalore (Upgrad Online),** *Certification on Business Analytics*
2022 – 2023 6 months

## WORK EXPERIENCE

**Nurture (AFS) Agtech Pvt. Ltd, GIS Engineer (ML-GIS Department)**
**Bangalore | May 2022 to Present**

• Working on Burn Detection, Tillage Detection, Tile Framework etc. and developing Kubernetes pipeline using different tools.

• Build full pipeline from raw MSI/SAR tile data to ingesting insights to DB, logging every step using PySpark and below cloud tools.

• Farm boundary detection using MRCNN (earlier), improved version with SAM-MRCNN using mapbox tiles.

• Working with Big Data using PySpark, AWS Athena Query (using python), AWS S3 storage, GitLab, Docker, Jenkins, AWS ECR (Docker Image Repo), Kubernetes CLI (kubectl), Spinnaker (Kubernetes 2.0), Kubernetes Dashboard, Logs etc. Prometheus, Grafana for Kubernetes Logs, Scheduling Kubernetes Jobs, Azure Databricks.

• Working with Geospatial Data and Sustain Projects using Python, Google Earth Engine and QGIS for Sustainability Projects.

• Building production-grade REST-like API endpoints using self-hosted **n8n Webhooks** and high-security automation workflows.

**SkyMap Global Pvt. Ltd, GIS Engineer**
**Kolkata | September-2021 to March-2022**

• Administrative Mapping Project: Village level Administrative Layout making, three tier Administrative Boundary creation and Database updating.

**Aabsys IT Pvt. Ltd, GIS / CAD Engineer**
**Bhubaneswar | March-2021 to September-2021**

• Navigation Mapping Project: City level Road and Local Administrative boundary extraction and validate using surveyed geotagged photos.
• Working with ArcGIS Python Toolbox, ArcGIS Modeler and Python to develop some tools that help co-workers to speed up the work.

## OTHERS TRAINING AND COURSES

**UCDEVIS University of California [Coursera Online]**
***Geographic Information System (GIS) Specialization***
2020-2021 [6 Months]

**IIRS-ISRO Outreach Programme,**
***Basics of RS, GIS and GNSS*** *(90 hours),* ***Overview of Geo-processing using Python*** *(13 hours,30 minutes)*
2020, 2021

ESRI Online Training [MOOC Courses], ***Do-It-Yourself Geo Apps*** *(4 weeks),* ***Cartography*** *(6 weeks),* ***Spatial Data Science*** *(6 weeks),* ***The Location Advantage*** *(6 weeks).*
2019, 2020

## SKILLS OR WORKING WITH TECHNOLOGIES BELOW:

ArcGIS & ArcGIS Pro,
QGIS, ENVI & ERDAS Imagine,
AutoCAD Map 3D,
SPSS, Advance Excel, Word (Docs),
PowerPoint (Slides),
Jira, Zoho – (Project Management Tool)
Sustainability Projects (Carbon Credit),
GeoServer, PostgreSQL (PostGIS),
Python, PySpark, Databricks (Python),
AWS S3, AWS Athena Query, AWS ECR,
Jenkins, GitLab, Kubernetes, Spinnaker,
Google Earth Engine (JavaScript),
R Programming Language,
Tableau, Power BI
SQL, TensorFlow, n8n

### QGIS Plugins

1. **MAS Spatial Analysis Tool:**
   Comprehensive DEM analysis toolkit with 45+ hydrological, geomorphometric, and stream network tools. Numba-accelerated for fast processing. **A Sets of QGIS Providers.**

2. **Open Geodata Browser:**
   Browse and download satellite imagery from STAC-compliant catalogs (Planetary Computer, AWS Earth Search, custom endpoints). **Satellite Data Downloading.**

3. **Advanced Map Downloader:**
   Advanced map download tool with multiple coordinate input types, canvas drawing, and georeferenced output formats.

4. **Raster Calculator:**
   A powerful map algebra raster calculator with professional-grade expression support.

5. **CSV Geometry Export:**
   Export vector layers as CSV files with geometry and customizable column names. **Creating QGIS Context menu and positioned to a desired position.**

6. **CSV Geometry Import:**
   Import CSV files with various geometry formats (WKT, WKB, GeoJSON, KML, Earth Engine, TopoJSON, X-Y coordinates).

7. **Advanced Selection Table:**
   Professional attribute table with two-tier selection (Cyan = Selected, Yellow = Highlighted) for advanced feature filtering and editing. **Creating secondary Attribute Table for Selection over selection.**

**Standford Online [Coursera Online]**, ***Machine Learning***
2019 – 2020

## PROJECTS OUTSIDE WORKING EXPERIENCE CLICK HERE FOR MORE

### Python (PyPi) Packages *(For all new packages, please visit: https://pypi.org/user/mirjanalisha/)*:

- **Open GeoData API** (`pip install open-geodata-api`):
  Open Geodata API provides seamless access to multiple open geospatial data APIs (Planetary Computer, Earth Search) with automatic URL management, intelligent filtering, and maximum flexibility for data reading, processing, downloading data with it's core STAC management.

- **PyClassStruct** (`pip install PyClassStruct`):
  A powerful CLI tool that converts simple Python scripts to well-organized class-based structures without changing logics.

### Complete Application *(For testing, please visit https://farmvision.onrender.com/)*:

- **FarmVision - Satellite Crop Analytics Platform** (`Build Using AI tools`):
  FarmVision is a lightweight, responsive web application for monitoring agricultural fields using satellite imagery and vegetation indices. It features a Python Flask backend that connects to a PostgreSQL database and serves raw GeoTIFF imagery directly to a modern, pure HTML/CSS/JS frontend.

### QGIS Plugins *(For all new plugins, please visit https://plugins.qgis.org/search/?q=Mirjan+Ali+Sha)* :

8. **Band Stacker:**
   A plugin to stack bands from selected raster layers.

9. **Clear Null Geometry:**
   This plugin deletes all NULL geometry from the shapefile. New users of QGIS easily creates NULL geometry to shapefiles when they trying to delete a geometry and facing difficulties to delete all of them easily, this plugin will help them to delete all NULL geometry from all selected layers. **Work without opening any dialog boxes.**

10. **CRS Finder:**
    Intelligent Coordinate Reference System finder and assign projection to the Layer on the fly. **Creating QGIS DockWidget and extensively used temp file and updating continuously on it.**

11. **Export 2KML:**
    Export multiple/single vector and raster layer/s into a single KML or KMZ with QGIS styling. And you have full options (all KML properties) for your output KML/KMZ. **Creating png preview files for raster data.**

12. **Mosaic Tool:**
    A plugin to mosaic bands from selected raster layers. **Creating QGIS Processing Tool and Tool Group.**

13. **Supervised Classifier:**
    Classify Raster Using Shapefile (.shp). A plugin to classify a selected raster file with reference. **Use external dependency like scikit-learn.**

14. **Unsupervised Classifier:**
    Plugin for unsupervised classification (clustering) of satellite images. **Use external dependency like scikit-learn.**

15. **XYZ/MGRS Coordinate Tool:**
    Advanced multi-format coordinate tool supporting XYZ tiles, MGRS, WKT, WKB, and GeoJSON with interactive navigation and visualization. **Adding Tool Pannels.**